

Frequency-scaled thermal-aware test scheduling for 3D ICs using machine learning based temperature estimation

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ABSTRACT

Three-dimensional Integrated Circuits (3D IC) are being looked at as an alternative in overcoming some critical challenges plaguing conventional 2D ICs. However, it faces some new challenges — some of them being thermal management and a complicated testing process. This work presents a frequency-scaled thermal-aware test scheduling algorithm for 3D ICs. The thermal awareness is provided by a proposed machine learning based predictive thermal estimation model for 3D ICs. The proposed test scheduling algorithm includes test clock frequency as an important parameter and scales the frequency to achieve lower test time. Experimental results for both the predictive thermal model and scheduling algorithm look promising, with improvements over existing works achieved in most of the cases. The thermal model estimates the temperature of a 3D IC with a high degree of accuracy but at a fraction of time of commonly used 3D IC thermal models, while the scheduling algorithm has been observed to have improved test time of different benchmark circuits up to around 60%. Experimental results show — that test time in terms of clock cycles is inconsequential because schedules with higher clock count may require less time due to frequency scaling.

1. Introduction

Three-dimensional Integrated Circuits (3D IC) have been instrumental in reducing signal delay, power consumption, footprint, and increasing throughput of modern Systems-on-chip (SoC) [5–8]. The improvements have been achieved by stacking multiple dies on top of one another with inter-die power routing, and communication implemented using Through Silicon Via (TSV) [5,9]. However, it gives rise to some additional challenges that must be mitigated to make the 3D IC technology viable for large-scale adoption. The stacking of dies leads to a longer heat dissipation path and consequently a higher operating temperature. Also, the upper dies of the stack are only accessible through the bottom-most die. This makes the testing procedure much more complicated than conventional 2D ICs, and so it requires sophisticated test procedures and scheduling algorithms.

High-volume post-production testing of ICs is complicated and time-consuming. Stacking of dies in 3D IC exacerbates the problem. Multiple dies requiring longer test processes ultimately increase the test cost. So, an optimized test schedule is necessary to minimize the test time of a 3D IC. However, generating a proper test schedule is a complicated task due to the large number of variables that must be considered. A 3D IC generally contains multiple modules, and each requires a

specific amount of time for its testing. All these modules must be accommodated in the test schedule at a proper point in time to keep the total test time low. Also, as the top dies are only accessible through the bottom-most die, the test data (patterns and responses) are transferred between the test pins and the modules in different dies with the help of test access mechanisms (TAM). However, TAMs are costly, so the number of test pins that can be utilized is also limited. This requires efficient use of the TAMs by the schedule, and they must be re-allocated to test other cores.

Modern scan-based test architecture greatly enhances the switching activity in the circuits under test (CUT), increasing the test mode power consumption. This, coupled with the fact that 3D ICs have a longer heat dissipation path, leads to thermal management issues. So, generating a test schedule that aims only to minimize the test time does not suffice. The test schedule needs to be thermal-aware in nature to ensure device safety. So it must be determined whether the temperature at any point during testing exceeds the safe thermal limit. This makes the task of generating the test schedule for 3D ICs even more complicated. So mathematical thermal models for 3D ICs are employed in parallel to guarantee a thermally safe test schedule. The thermal models estimate

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the temperature of a 3D IC according to a schedule, and necessary steps are taken to tweak the schedule to make it thermally compliant.

Most mathematical thermal models like Hotspot [3], 3D-ICE [10], Comsol [4] employ finite element modeling (FEM) that are resource-intensive in nature. They require a substantial amount of time to calculate the temperature of a 3D IC. So, when they are employed with test schedule generating algorithms, the whole process becomes quite complicated and time-consuming. This problem is somewhat mitigated by employing steady-state thermal solvers, which take comparatively less time but can only calculate the steady-state temperature of a 3D IC. Transient thermal solvers can calculate the temperature at any point in time but require much more resources and computation time.

Due to the heightened switching activity in scan-based testing, the test clock frequency is often reduced several-fold compared to the operational clock frequency to keep the test mode power dissipation in check. Test clock frequency directly affects power consumption and consequently the temperature of a 3D IC. It also affects the amount of time a 3D IC spends on the test equipment thus, determining the test time and cost. Lower test clock frequency may reduce the power consumption, but it increases the test time. However, increasing test clock frequency may reduce test time but may render the test schedule thermally unsafe. So, the test schedule must run at an optimized test clock frequency in order to further reduce the test time and cost.

Considering the test challenges of modern 3D ICs, this work aims to develop test schedules for 3D ICs that are thermal aware in nature while minimizing the test cost. The temperature and power consumption of a 3D IC is directly co-related. Existing FEM-based mathematical thermal models are slow due to the large number of elements (for better accuracy) and the matrix inversion operation required in the process. So, a machine learning (ML) based approach is developed to estimate the temperature of 3D ICs. ML-based techniques are pretty fast if we discount the training time (which is required only once). It is also accurate to a high degree with proper training. The ML-based thermal model is employed with the 3D IC test scheduling algorithm to provide thermal awareness. The heuristic nature of the proposed test scheduling algorithm combined with the ML-based thermal evaluation makes the test schedule generation quite fast, accurate, and thermally safe. The scheduling algorithm also incorporates test clock frequency to employ frequency scaling, which further helps to reduce the test cost. Thus, the main contributions of the paper can be summarized as:

1. ML-based thermal estimation model is developed for 3D ICs.
2. A heuristic test scheduling algorithm is developed for 3D ICs.
3. Frequency scaling is employed to reduce 3D IC test cost.

The rest of the paper is organized as follows. Section 2 reviews the earlier relevant works, followed by the motivation behind this work and the problem formulated in Section 3 and Section 4, respectively. Section 5 explains the methodology adopted in this work. Results and discussions are presented in Section 6. Section 7 presents the conclusion to this work.

2. Literature review

Considerable work has been done regarding thermal modeling and test scheduling of 3D ICs. In this section, we review the existing works that are relevant to the work presented here. Thermal management issue has been a significant challenge for 3D ICs since their inception. Consequently, enormous effort has been put into accurate thermal modeling of 3D ICs. With the stacking of multiple dies, the vertical heat dissipation path in a 3D IC is long and heterogeneous. Thermal models for conventional 2D ICs fail to emulate the complicated nature of heat flow in 3D ICs. So, dedicated 3D IC thermal models are required.

The most popular and widely used thermal models for 3D ICs are Hotspot [3], 3D-ICE [10], and Comsol [4]. A majority of existing literature makes use of these models. These models can be classified as finite

element solvers where the structure of the 3D IC is logically divided into small cells or elements. The time domain is also discretized. These models are highly accurate in estimating the thermal profile of a 3D IC. However, due to the large number of elements created in the process, these models can be slow and resource-intensive.

Several other mathematical thermal models adopt different approaches to estimate the temperature of 3D ICs. Ayala et al. [8] present a 3D IC thermal model analogous to modeling electric circuits where the temperature is analogous to voltage and heat flow to electric current. Im et al. [11] utilize the Joule heating formula to determine the thermal profile of 3D ICs. Skadron et al. [12] estimate the temperature based on thermal resistance at different points in the IC. Shih et al. [1] use a one-dimensional thermal model in test scheduling where lateral heat flow is neglected due to the low thickness of 3D IC dies. This speeds up the calculation sacrificing accuracy. However, all these models solve for the temperature of a 3D IC by numerical analysis, which makes them comparatively slow.

Some innovative techniques have also been proposed to speed up the thermal estimation process. Karmakar et al. [13] store the temperature of individual modules in a database and applies the superposition principle to quickly calculate the temperature for different combinations of cores of 2D SoCs. This leads to fast temperature calculation but may require a substantial database size for implementation in 3D ICs with a large number of cores. Samal et al. [14,15] use a non-linear regression-based technique to calculate the temperature for floorplanning of monolithic 3D ICs. However, the input parameters are derived from basic parameters and need additional computation time. It also changes the input parameter values depending on the position of the target cell. Due to this, the computation time required to generate the thermal profile of the entire 3D IC may increase. In this work, we present a ML-based thermal model where the power dissipation of cells are fed in directly as input. This enables the model to estimate the temperature of the cells all at once thereby reducing computation time.

Chatterjee et al. [2] adopt a linear regression-based approach using power consumption of the dies of a stacked 3D IC to estimate its temperature, which is used for test scheduling of 3D ICs. Such a low number of input parameters makes the thermal estimation very quick but sacrifices accuracy. Other existing test scheduling algorithms can be broadly clustered into two groups that employ either heuristic or integer linear programming (ILP). Some works also employ meta-heuristics like simulated annealing (SA) [1] or particle swarm optimization (PSO) [13,16] for test scheduling of 3D ICs. The test application scheme has a significant impact on the temperature of an IC which may introduce some additional faults. Xiang et al. [17] studies the impact of test application on temperature which may introduce additional path delay leading to an overkill during small delay defect (SDD) testing.

Heuristic test scheduling algorithms [2,18–20] are fast and produce excellent results. Nevertheless, due to the always greedy approach of heuristics, sometimes the results can be poor. Meta-heuristic algorithms like SA used by Shih et al. [1] or PSO used by Karmakar et al. [13] employed for test scheduling can produce better results as they can be prevented from getting stuck in local optima. Hence, their solutions are more optimized and closer to the best solution. However, meta-heuristics can be slow as they may require multiple runs and are resource-intensive. Wu et al. [21] use ILP for test scheduling of 3D ICs, which is also thermal-aware. The use of ILP and mathematical thermal model produces very good and accurate results, but it requires the most time. However, all the test scheduling algorithms calculate test time in clock cycles and do not consider test clock frequency. Ling et al. [22] exploit the advantages of dynamic voltage and frequency scaling (DVFS) for testing scheduling of 2D SoCs under thermal constraint and achieve around 16% reduction in test time. However, in 3D ICs with stacking of dies and longer heat dissipation path, the advantageous effect of scaling on test time can be more pronounced. Xiang et al. [23] presents a unicast based multicast thermal-aware test

scheme for 3D NoCs where the temperature is reduced by avoiding sending test packets to thermal hotspot regions. The test cost is also affected by the number of TSVs used for testing. Cheng et al. [24] present a TSV minimization technique that reduces TSV count without negatively affecting the test time. Some prior works [21,25,26] put a TSV constraint on the testing scheme to keep the TSV count in check.

3. Motivation

The motivation behind this work arises from the complicated nature of testing 3D ICs. The objective is to generate safe test schedules for 3D ICs to reduce testing time as much as possible. To ensure safety, thermal validation of the generated test schedule is essential. Also, due to a several-fold increase in power consumption during testing, the test clock frequency is significantly reduced compared to operational clock frequency. However, both the Automatic Test Equipment (ATE) on which the IC is tested and the 3D IC itself are capable of operating at much higher frequencies. Almost all test scheduling algorithms in literature neglect this aspect and present their test times in terms of clock cycles. However, as we will see in later sections, accommodating the test clock frequency in the scheduling algorithm can significantly reduce the absolute test time (in seconds) of the 3D ICs.

All the 3D IC test scheduling algorithms in the literature are either thermally oblivious or aware in nature. Thermally oblivious test schedules generally produce better results in terms of test time as there are no thermal constraints placed on them. However, on validation, these schedules can be rendered unsafe and require further reduction in test clock frequency, ultimately increasing the test time. Thermally aware test schedules are safe, but the thermal validation is done with power values at a definite test clock frequency. The motivation of this work is derived from initial test results showing that tweaking the test clock frequency can significantly alter the schedule and greatly improve the test time of 3D ICs while ensuring thermal safety.

The time required to generate the test schedule is also of particular concern. With many variables weighing in on the test time, finding an optimized test schedule can be a time-consuming process. Simultaneously, in thermal-aware algorithms, analytical thermal models are employed to check the validity of the generated schedules. Most of these thermal models divide the 3D IC into a finite number of elements/cells. Generally, a heat flow matrix is formed in these models of the order of the number of elements. The temperatures of the elements are obtained by vector multiplication of the inverse of this matrix with the power distribution vector. This is a time-consuming process since the computational complexity of matrix inversion is directly proportional to the cube of the order of the matrix [27].

Consequently, the test schedule generation time worsens due to the large number of elements utilized in these models for higher accuracy. Such disadvantage is the primary motivation behind this work employing machine learning to predict the temperature of a 3D IC under any test schedule. Machine learning can be slow initially due to the requirement of collecting training data. However, once sufficient data is gathered, and the model trained properly, it can be quite fast and re-used for other 3D ICs (under some constraints) while providing respectably high accuracy.

4. Problem formulation

The total test time of a 3D IC depends on the test time of individual cores. The test time of the cores depend on multiple factors such as the number of scan chains, length of the scan chains, TAM allocation. Wider TAM allocation reduces the test time of a core, but up to a certain limit. The test time of the cores in this work is calculated according to the method presented by Marinissen et al. [28]. Thus, the total time of testing a 3D IC is reduced by testing the cores in parallel. However, excessive parallelism in some instances may increase the test time and lead to higher power consumption, violating thermal safety.

Thus, a combination of parallel and serial testing can be adopted for an optimized result.

So, given a 3D IC with (i) Maximum TAM width (W_{max}), (ii) D number of dies, (iii) N number of cores, (iv) Test time of individual cores under various TAM widths, (v) Power consumption of the cores at a definite test clock frequency, and (vi) Placement of cores in different dies, generate an optimized test schedule for a 3D IC at a particular test clock frequency under maximum TAM width of W_{max} to minimize the absolute test time while ensuring the thermal safety.

5. Methodology

This section explains in detail the techniques adopted to implement this work. The proposed test scheduling algorithm is heuristic, session-based, and thermal-aware in nature. Since the scheduling algorithm uses the ML-based temperature prediction scheme in finding thermal-aware schedules, the thermal estimation model is described in the first part of this section. The first part deals with the data collection process, training methodology, and cost function for the ML-based thermal model. The second part explains the test scheduling algorithm in detail, followed by complexity analysis and an example with a 3D IC benchmark.

5.1. Thermal estimation model

In a 3D IC, dies are stacked on top of one another, which leads to longer heat dissipation paths. The heat generated in a die must flow through all the dies above it to reach the heat sink. This means a die must not only transfer its heat to the die above it, but also the heat generated in all the dies below. Given that the thermal conductivities of the dies are fixed by the material properties, the temperature difference between the dies must increase to accommodate the heat from the bottom dies. This leads to high temperature in 3D ICs.

Most 3D IC thermal models adopt Finite Element Modeling (FEM) analysis to calculate the temperature. The whole 3D IC stack is logically divided into small elements/cells, as shown in Fig. 1(a) [1]. Here Shih et al. present a one-dimensional thermal model for 3D ICs, where heat flow is considered only in the vertical direction. The lateral heat flow is neglected because it is considered insignificant compared to the vertical heat flow as the thickness of the dies in a 3D IC is very less. So, the temperatures of each stack of cells are solved independently. Although one-dimensional models are fast, they can be inaccurate due to the omission of lateral heat flow. The electrically analogous thermal resistance model is shown in Fig. 1(c) [1]. The temperature T_0 is calculated in [1] as in Eq. (1), where P_0 to P_2 and R_0 to R_2 are power dissipations and thermal resistances of the corresponding cells shown in Fig. 1(b) respectively, $T_{ambient}$ and R_b is the ambient temperature and thermal resistance respectively, and T_0 is the temperature of the bottom-most cell.

$$T_0 = P_0 \times (R_0 + R_1 + R_2 + R_b) + P_1 \times (R_1 + R_2 + R_b) + P_2 \times (R_2 + R_b) + T_{ambient} \quad (1)$$

The temperature of a 3D IC is directly proportional to the power consumption and obeys the principle of superposition [29]. These types of relations can be well-modeled by linear regression equation. So, Eq. (1) can be approximated through a multi-variable linear regression model as expressed in Eq. (2), where f_1 to f_n are the input variables (in this case P_0 to P_2), θ_0 to θ_n are the regression model parameters, and $\bar{T}_0$ is an approximation of T_0 .

$$\bar{T}_0 = \theta_0 + (f_1 \times \theta_1) + (f_2 \times \theta_2) + \dots + (f_n \times \theta_n) \quad (2)$$

The main challenge is to find the proper values of the regression model parameters, such that the error between the approximate value ($\bar{T}_0$) and the actual value (T_0) is minimized. This work employs machine learning (ML) in this endeavor. For ML to work, proper data must be collected from numerous 3D ICs and the model trained properly to find the best regression parameter values. Let us now look at the data collection process followed by the training methodology.

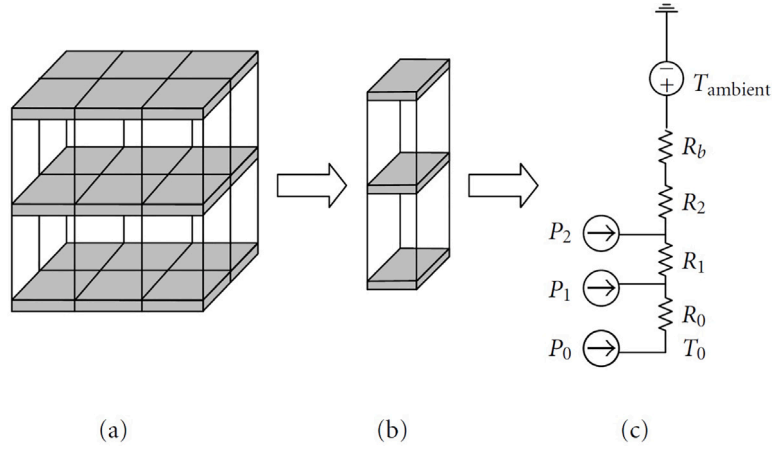


Fig. 1. (a) A 3×2 array of cell stacks. (b) A cell stack. (c) Thermal resistance model of a cell stack. [1].

5.1.1. Data collection

Like FEM thermal models, this work divides the 3D IC into logical elements/cells. We consider a 5-die 3D IC, and each die is divided into 10×10 cells. Like other existing thermal models in the literature, this work considers the package side to be adiabatic, and all heat is dissipated to the environment through an ambient thermal resistance connected to the top of the stack. Numerous arbitrary 3D ICs with varying number of cores and dies, core sizes, core placement, power dissipation are generated and mapped into this 5-die 3D IC. 3D ICs with fewer dies are mapped by assigning the power dissipation of non-existing dies (package side) to zero. For example, a 3-die 3D IC is mapped by considering the power dissipation of the two bottom-most dies to be zero.

Since the package side is considered to be adiabatic, this approximation does not affect the steady-state temperature of the dies. In reality, the package side is not adiabatic, but a small amount of heat does flow through this side. However, it is significantly less compared to the top side, where the heat sink is connected. The small amount of heat flowing through the package side only brings the temperature of the 3D IC down, thus, not violating the thermal safety guaranteed by this type of model, which considers the package side to be adiabatic. However, it leads to a very small overestimation of temperature, which may, in turn, sacrifice some test time in schedules employing these models.

Each randomly generated 3D IC is simulated for thermal estimation using the academic thermal estimation tool Hotspot [3]. The temperature collected from Hotspot is considered to be the actual temperature of a 3D IC for the purpose of this work. Thermal simulation in Hotspot is performed using 256×256 grid model with lateral heat flow switched on. Since the lateral heat flow is considered while estimating the actual temperature, the power dissipation of all the cells must be considered for approximating the temperature of each cell. The temperature of all the cells calculated from Hotspot is stored along with the maximum temperature of the 3D IC. Fig. 2 shows the table schema of the collected data, where each row is a randomly generated 3D IC mapped into the 5-die 3D IC, P_1 to P_{500} represents the power dissipation of the cells of the 5-die 3D IC, T_1 to T_{500} are the corresponding temperatures gathered from Hotspot, and T_{max} is the maximum temperature.

The whole process is run iteratively, generating a random 3D IC, mapping it into the 5-die 3D IC, running a thermal simulation of the generated 3D IC, power and temperature information collected and stored. The tasks repeat until a sufficient number of sample data has been gathered. The process is summarized in Fig. 3 flowchart.

5.1.2. Training

Proper training is required for high accuracy of the ML system. Here, in regression-based learning, training refers to finding the values of the regression parameters ($\theta \rightarrow \theta_0$ to θ_n) expressed in Eq. (2). We train

P_1	P_2		P_{500}	T_1	T_2		T_{500}	T_{max}
.....								
.....								
.....								

Fig. 2. Structure of the collected data as stored for the purpose of training.

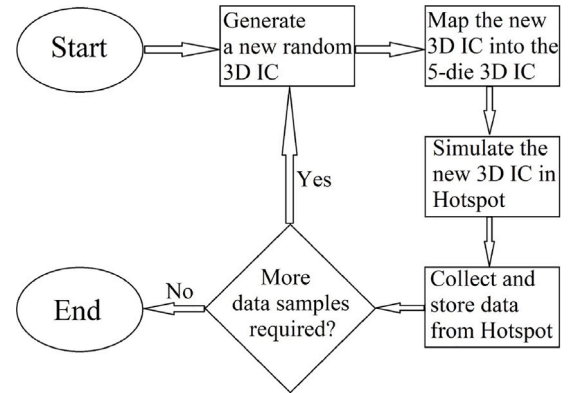


Fig. 3. Data collection process flowchart of generating random 3D ICs, simulating in Hotspot and collecting power and temperature data.

the system using the data gathered from over ten thousand randomly generated 3D ICs. Required data is extracted, stored in variables, fed as input to the system to predict the temperatures of the 3D ICs with the aim to minimize prediction error.

The power consumption and temperature data of thousands of random 3D ICs are stored in the data set. For the purpose of training 80% of the 3D ICs are selected randomly. The remaining 20% are kept for testing and validation. The regression parameters are calculated by the equation shown in Eq. (3). The regression parameter (θ) has a dimension of 500×500 . Each column in theta represents the regression parameters of each cell in the 3D IC. Temperature of each cell depends on its power consumption and the power consumption of every other 499 cells. So, each column in θ has 500 rows, and there are 500 columns — one each for every cell.

$$\theta = ((P' \times P)^{-1} \times P') \times T \quad (3)$$

Here, P and T are the matrices containing power consumption and actual temperature respectively of all cells of all the random 3D ICs

in the training set. Once the regression parameters are calculated, the model is trained. Eq. (3) is used in training of the model due to the relatively low feature size in this problem. However, this equation becomes problematic for larger feature sizes due to matrix inversion required in this equation. In such cases, a heuristic algorithm like gradient descent can be used. Using θ calculated from Eq. (3), we can now predict the temperature of any similar 3D IC, as shown in Eq. (4).

$$\bar{T} = (P_{distribution} \times \theta) + T_{ambient} \quad (4)$$

Here, $\bar{T}$ represents the predicted temperatures of all the five hundred cells of the 3D IC. $P_{distribution}$ is the array containing power consumption of all 500 cells of the 3D IC and θ is the regression parameter calculated from Eq. (3). The maximum temperature of the 3D IC can be calculated as in Eq. (5).

$$\bar{T}_{max} = \max(\bar{T}) \quad (5)$$

The process of selecting 80% of 3D ICs for training, followed by computation of Eqs. (3)–(5) is repeated for over ten thousand times and the θ producing minimum prediction error over the test set is kept for further use. The peak temperature of a 3D IC can also be predicted directly from the actual peak temperatures of the random 3D ICs of the training set. Eq. (6) shows the calculation of the regression parameters, where θ_{max} is the regression parameter for calculating the peak temperature of a 3D IC directly and T_{max} is the actual peak temperatures of the random 3D ICs in the training set.

$$\theta_{max} = \left((P' \times P)^{-1} \times P' \right) \times T_{max} \quad (6)$$

The peak temperature of a 3D IC can be predicted directly using θ_{max} as shown in Eq. (7). Both the processes of prediction using θ and θ_{max} have been tested for this work. However, it is found that using the former leads to higher prediction accuracy while using the latter is faster by an order of three.

$$\bar{T}_{max} = (P_{distribution} \times \theta_{max}) + T_{ambient} \quad (7)$$

Further refinement can be made to the model parameter values to improve accuracy by employing Particle Swarm Optimization (PSO) [30]. By PSO, we are searching for better values of θ . Five hundred particles are introduced in the search space with random initialization. Since the feature size is 500, the search space includes 500 dimensions. PSO is run iteratively for each of the 500 cells in the 5-die 3D IC.

However, PSO suffers from premature convergence and the same was noticed in this work. So, one of the particles is initialized with values of θ_j , when the current PSO iteration is running for the j th cell. Concatenating horizontally the values of $\theta_j, \forall j \in \mathbb{Z} | 1 \leq j \leq 500$ in order, gives the improved values of θ . It is however observed, though PSO improves the regression parameters, the improvement is minimal.

The entire training process, both Eq. (3) to Eq. (7) and the PSO, uses the Mean Square Error (MSE) cost function metric. The Mean Square Error (MSE) cost function can be calculated as shown in Eq. (8), where T_i^j and $\bar{T}_i^j$ are the actual and predicted temperatures respectively of the j th cell in the i th 3D IC of the training set. M is the number of samples in the test set. The MSE is also employed to select the best θ and θ_{max} over thousands of iteration. Other cost functions like Sum Square Error (SSE) and Mean Square Relative Error (MSRE) have also been tried, but MSE is found to be producing the best results.

$$MSE = \frac{1}{M} \sum_{i=1}^M (T_i^j - \bar{T}_i^j)^2 \quad (8)$$

5.2. Test scheduling

The 3D IC test scheduling algorithm presented here is session-based. A session is referred to as multiple cores being tested in parallel, and no additional cores are scheduled until all the present cores are finished testing. The scheduling algorithm is designed to reduce test time while

keeping under the thermal safety limit. So, given a number of cores in a session, let us describe a process of assigning test pins to the cores, so as to reduce the test time of the cores in the session.

Algorithm 1 describes the process of assigning test pins to the cores in order to minimize test time of the session. Firstly, only one pin is assigned to each core of the session, and their test times are calculated. The cores are sorted in descending order of their test times with conflicts being resolved arbitrarily. Next, one extra test pin is assigned to the first core in the sorted list (core with largest test time), and if its test time reduces, the list is sorted again in descending order of their test times, else one more test pin is allocated again to the largest core. The process repeats until all the test pins have been allocated.

Algorithm 1: getPinAssignment()

input : List S of N_s cores in the session.
 Total TAM width (W_{max}).
 Test time (TT) of each core in S under various TAM width.
output : Pin assignment of each core in S .

- 1 Assign 1 test pin to each core in S ;
- 2 Calculate test times of cores in S ;
- 3 sort(S); /*Descending order of TT for assigned test pins */
- 4 $usedPins \leftarrow length(S)$;
- 5 $avlblPins \leftarrow W_{max} - usedPins$;
- 6 $sessionTime \leftarrow$ Test time of 1st core in S ; /*1st core is the largest core*/
- 7 **while** $avlblPins > 0$ **do**
- 8 Assign 1 more pin to 1st core in S ;
- 9 Calculate test time of the 1st core in S ;
- 10 $newSessionTime \leftarrow$ Test time of 1st core in S ;
- 11 $usedPins \leftarrow usedPins + 1$;
- 12 $avlblPins \leftarrow W_{max} - usedPins$;
- 13 **if** $newSessionTime < sessionTime$ **then**
- 14 $sessionTime \leftarrow newSessionTime$;
- 15 sort(S); /*Descending order of TT*/

While Algorithm 1 describes the pin assignment to the cores of a session, Algorithm 2 explains how cores are added to a new session. A list of all the unscheduled cores are sorted in non-increasing order of their test times with each core provided the full TAM width (W_{max}). The largest core (in terms of test time) is added to the new session and the temperature is calculated. If the temperature exceeds the safety limit (T_{limit}), the core is removed, and the session is aborted. This is done because there is at least one core which cannot be scheduled while not violating the thermal safety. Such situation requires reduction of test clock frequency, thus, reducing power dissipation of the core and hence, the temperature.

If the first core of the session does not violate the thermal safety, subsequent cores are added to the session based on two conditions. First, addition of a new core must not violate the thermal safety. Second, after new pin assignment to the cores, the increase in test time of the session must not exceed the test time of the newly added core when tested with full TAM width. If the increase in test time of the session is greater than test time of the new core under W_{max} , it is beneficial to keep the session unaltered and schedule the new core in the next session. The process repeats until all the unscheduled cores are checked whether it can be added to the session satisfying the above two conditions.

To access the non-bottom dies for testing, TSVs are required. However, TSVs in 3D IC are limited due to their area overhead. The total number of TSVs required by the i th core of the 3D IC is $W_i L_i$, where W_i is the assigned TAM width to the core and L_i is the die number. The number of TSVs required in a session is equal to the sum of all the TSVs used by the cores of that session. The TSV count of the entire schedule is calculated as the maximum number of TSVs utilized over all the sessions. A TSV constraint can be placed like the temperature

limit in Algorithm 2 to reduce the number of TSVs that can be used by the schedule.

Algorithm 2: getSession()

input : List U of unscheduled cores in the 3D IC.
 Total TAM width (W_{max}).
 Test times (TT) of cores in U under W_{max} .
 Power dissipation of the cores (P_{scale}).
 Thermal safety limit T_{limit} .
 TSV limit TSV_{limit} .

output : Cores of the new session with pin assignment.

```

1 Begin new session ( $S$ );
2 sort( $U$ ); /*Descending order of TT for  $W_{max}$  */
3  $S.add(U_1)$ ; /*First core in  $U$  */
4  $TAM(U_1) \leftarrow W_{max}$ ;
5  $sessionTime \leftarrow TestTime(S)$ ;
6  $Temp \leftarrow CalculateTemperature(S, P_{scale})$ ;
7 if  $Temp < T_{limit}$  then
8   for  $i \leftarrow 2$  to  $length(U)$  do
9     if  $length(S) < W_{max}$  then
10        $S.add(U_i)$ ;
11        $Pins \leftarrow getPinAssignment()$ ;
12        $Temp \leftarrow CalculateTemperature(S, P_{scale})$ ;
13       if  $Temp > T_{limit}$  or
14          $TestTime(S) - sessionTime > t_i$  then
15         /* $t_i \Rightarrow$  TT of  $i^{th}$  core in  $U$  with TAM  $W_{max}$  */
16          $S.remove(U_i)$ ;
17       else
18          $sessionTime \leftarrow TestTime(S)$ ;
19          $TAM(S) \leftarrow Pins$ ;
20 else
21    $S.remove(U_1)$ ;

```

Consequently, using the session formation process described in Algorithm 2, the entire test schedule for the 3D IC is generated as presented in Algorithm 3. First, a fully serial test schedule is generated providing each core the maximum TAM width (W_{max}), and testing each one after another. For now, it is considered the best test schedule. Subsequently, the serial schedule is improved upon by moving cores into sessions in which multiple cores are tested in parallel. The process of forming new sessions and allocating cores to new session repeats until there are no unscheduled cores left. Thus, a new schedule is formed.

However, schedules are generated for various test clock frequencies by sweeping the frequency scale (f_{scale}) up to five times of the base test clock frequency. The best schedule (in terms of test time) is kept along with the frequency scale at which it was generated. As dynamic power dissipation is proportional to frequency, the cores' power dissipation is recalculated for each new frequency scale. This tweaks the sessions, and consequently, the schedule at each new frequency scale. However, test time can be determined in two ways. First, test time in terms of clock cycles, and second, absolute test time in seconds, taking into account the frequency scale. The following Results section shows the best test schedules using both ways for different 3D ICs. It is often observed that the best test schedule in terms of clock cycles may not have the best absolute time (in seconds).

In some cases, the actual temperature of the schedule may exceed the thermal safety limit because the proposed thermal estimation technique is predictive in nature. So, the generated test schedule must be verified for thermal safety using a deterministic mathematical thermal model. If the schedule is found to have violated the thermal safety limit by the mathematical model, the frequency scale (f_{scale}) is adjusted according to Eq. (9) in order to bring the temperature (T_{max}) under the safety limit.

$$f_{scale_{new}} = f_{scale} \times \frac{(T_{limit} - T_{ambient})}{(T_{max} - T_{ambient})} \quad (9)$$

Algorithm 3: schedule()

input : List C of N cores of the 3D IC.
 Total TAM width (W_{max}).
 Power dissipation of cores (P) at base test clock frequency.

output : Test schedule of the entire 3D IC.
 Frequency scale.

```

1 Begin new schedule ( $bestSchedule$ );
2  $bestScheduleTime \leftarrow 0$ ;
3 for  $i \leftarrow 1$  to  $N$  do
4   /*Generate fully serial schedule*/
5    $TAM(C_i) \leftarrow W_{max}$ ;
6   Begin new session( $S$ );
7    $S.add(C_i)$ ;
8    $bestSchedule.add(S)$ ;
9    $bestScheduleTime \leftarrow bestScheduleTime + t_i$ ;
10  $f_{scale} \leftarrow 0.05$ ;  $bestF_{scale} \leftarrow 0.05$ ;
11 while  $f_{scale} \leq 5$  do
12    $U \leftarrow C$ ; /*  $U$  tracks unscheduled cores */
13   Begin new schedule( $X$ );
14    $scheduleTime \leftarrow 0$ ;
15    $P_{scale} \leftarrow f_{scale} \times P$ ;
16   while  $!U.isEmpty()$  do
17     Begin new session( $S$ );
18      $S \leftarrow getSession()$ ;
19     if  $S.isEmpty()$  then
20        $X.empty()$ ; break;
21     else
22        $X.add(S)$ ;
23        $U.remove(S)$ ;
24        $scheduleTime \leftarrow scheduleTime + TestTime(S)$ ;
25   if  $X.isEmpty()$  then
26     break;
27   else
28     if  $scheduleTime \leq bestScheduleTime$  then
29        $bestSchedule \leftarrow X$ ;
30        $bestScheduleTime \leftarrow scheduleTime$ ;
31        $bestF_{scale} \leftarrow f_{scale}$ ;
32    $f_{scale} \leftarrow f_{scale} + 0.05$ ;

```

5.3. Complexity analysis

The computation time of the presented technique depends on the time complexity of the thermal estimation and the test scheduling algorithms. For thermal estimation, the active cores need to be mapped to the 5-die 3D IC. If there are N active cores, the time complexity of mapping is $O(N)$. However, the calculation of temperature using Eq. (2) has fixed time complexity as the number of cells in the 5-die 3D IC is fixed.

Algorithm 1 assigns test pins to the cores of a session. First, it assigns one test pin to each core and sorts them in terms of their test times. This requires $O(N_s^2)$ time, where N_s is the number of cores in the session. However, it keeps on adding one test pin to the largest core and sorting, until there are no available test pins. So, all the test pins are allocated in $O(W_{max})$ time and the previously largest core can be insert sorted into the already sorted list in $O(N_s)$ time. Hence, the combined time complexity is $O(W_{max}N_s)$. However, in most cases, W_{max} will be greater than N_s . So, the overall worst-case time complexity can be generalized as $O(W_{max}N_s)$.

In Algorithm 2, each unscheduled core is checked once if it can be added to the new session. So, its worst-case time complexity is $O(U)$, where U is the number of unscheduled cores. In Algorithm 3, first, a fully serial test schedule is generated whose time complexity is $O(N)$, where N is the number of cores in the 3D IC. As sweeping the frequency scale needs a fixed time, worst-case time complexity is achieved when each session contains a single core (which means a serial schedule).

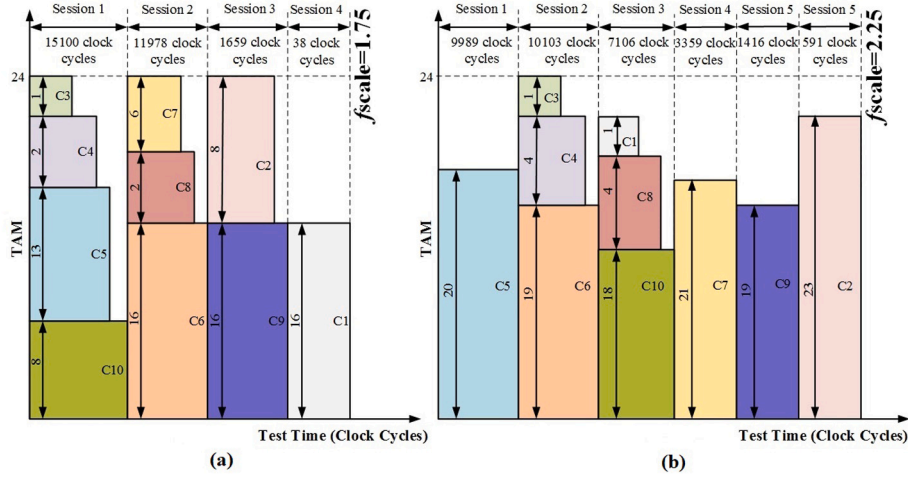


Fig. 4. Generated thermal-aware test schedule for d695 placed in 4-die 3D IC for minimized test time in terms of (a) Clock count (b) Absolute time.

Hence, the overall worst-case time complexity of Algorithm 3 is also $O(N)$.

5.4. Example

Consider the ITC'02 benchmark circuit - d695, placed in a 4-die 3D IC with cores placed in dies as described in [18]. The power dissipation of the cores is calculated at a base test clock frequency of 120 MHz at 45 nm technology node [31]. The thermal safety limit is set to 90°C with a maximum TAM width (W_{max}) of 24. Figs. 4(a) and 4(b) show the generated test schedule of the 3D IC for minimum test time in terms of clock cycles and absolute time, respectively.

Initially, the fully serial schedule is generated, giving each core the full TAM width (W_{max}) and sorted in non-increasing order of their test times. Since core C5 is the biggest core, it is added to the first session. In the schedule presented in Fig. 4(a), core C6 cannot be added to the first session despite being the next biggest core because the thermal safety limit gets violated. Consequently, C10 is added to the current session, as it neither violates the thermal safety limit, nor it increases the session time beyond its own test time in the fully serial schedule. Similarly, C4 is added, but not C8. Although C8 does not violate the thermal limit, it increases the session time by more than its own test time in the serial schedule. The idea behind this logic is that the core C8 will contribute more to the test time by adding it to the current session than placing it in the next session, even if it is the only core left. Finally, only one more core (C3) could be added to the first session among all the unscheduled cores satisfying the two conditions.

The test clock frequency scale ($fScale$) for the lowest clock count in test time (28775 clock pulses/0.137 mS) is observed at 1.75 (Fig. 4a). Though the schedule presented in Fig. 4(b) has a higher clock count (32564), yet it takes less amount of time (0.106 mS) because of a higher $fScale$ of 2.55. Both of these schedules are thermal-aware in nature, but they show that minimizing the clock count of a test schedule does not necessarily guarantee minimum test time on the ATE. The absolute test time of a thermal-aware schedule depends highly on the cores' power dissipation and consequently on the test clock frequency. However, most established works in literature emphasize reducing the clock count, neglecting the power dissipation factor. The test times of d695, both in terms of clock count and absolute time for different $fScale$ values are presented in Fig. 5 for better clarity.

6. Results and discussion

This section presents and analyzes the implementation results of techniques and algorithms discussed in the preceding section and realized on ITC'02 benchmark circuits [32]. The work is implemented using

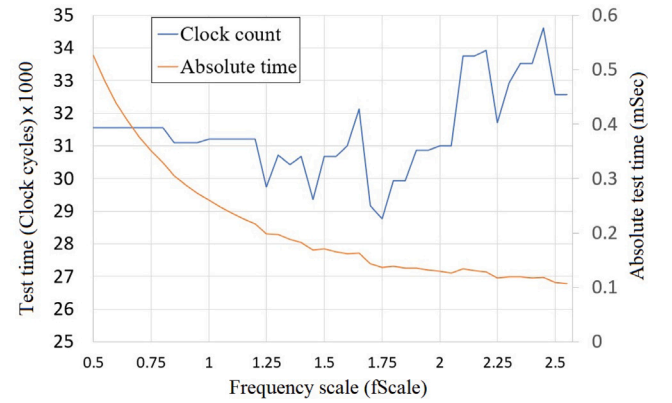


Fig. 5. Test time (clock count and absolute time) of d695 in a 4-die 3D IC for various test clock frequency scales ($fScale$).

Hotspot 6.0 [3], Java 7, Matlab 2015 (Linux environment), and run on a 2.4 GHz dual-core Intel Xeon processor with a primary memory of 8 GB. Results of several ITC'02 benchmark circuits are provided mapped to 3D ICs with floorplans and power dissipation values estimated for 45 nm technology node at a base test clock frequency of 120 MHz [31]. Table 1 lists the values of different system and environment variables used during the implementation of this work. The presented methodology can also be replicated for other values of these variables.

This section is further divided into two sub-sections. The results related to the accuracy of the predictive thermal estimation model is presented in the first sub-section, while the second shows the test scheduling algorithm results.

6.1. Thermal estimation

The proposed thermal estimation model is based on machine learning which require a large training data set. Over ten thousand random 3D ICs with varying floorplans and power dissipation values are simulated with the help of Hotspot 6.0 for the purpose of training the predictive thermal estimation model. For better accuracy, Hotspot is used in grid mode with lateral heat flow switched on. A large grid size of 256×256 is used to maximize accuracy. However, other thermal models or temperature data from an actual 3D IC can also be used for the purpose of training. The proposed thermal model presented here improves upon the predictive thermal model presented by Chatterjee et al. [2]. Table 2 compares the accuracy of the proposed thermal model (Celsius scale) with that of [2] over the entire data set.

Table 1

Variables and their values used for the implementation this work.

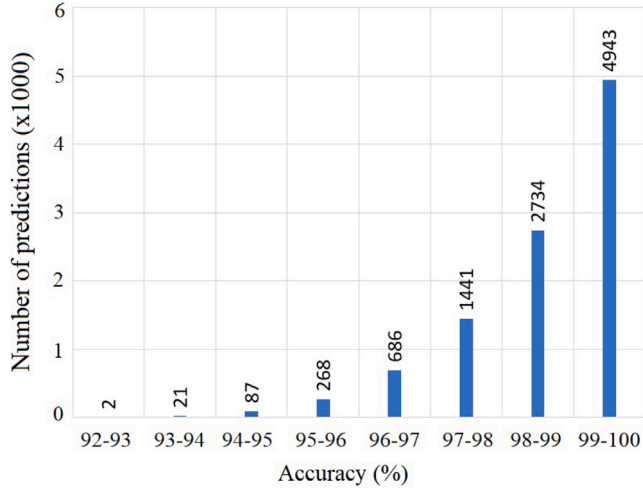
Environment variables		System variables	
Ambient temperature	27 °C	Silicon thermal conductivity	100 W/mK
Safe thermal limit	90 °C	Package thermal resistance	Adiabatic
Thermal resistance to ambient	0.5 K/W	Size of die	10 mm × 10 mm
–	–	Thickness of die	50 µm

Table 2

Accuracy of the proposed predictive thermal model compared to the model presented in [2].

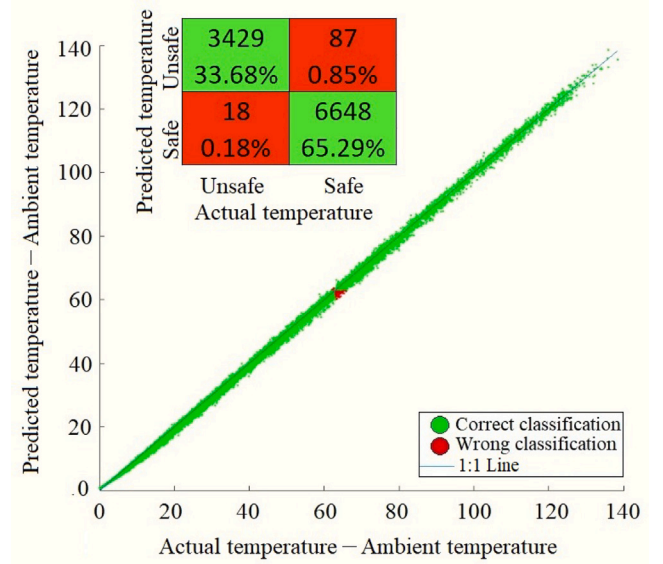
Thermal model →	Chatterjee et al. [2]	Proposed model
Minimum accuracy (%)	48.36	92.28
Average accuracy (%)	93.38	98.63
Maximum accuracy (%)	99.99	99.99
Standard deviation	6.5	1.18
Modal group	99%–100%	99%–100%

The frequency distribution of the accuracy is also an important parameter in estimating the performance of the predictive model. Fig. 6 shows the frequency distribution in accuracy over the entire data set. It is to be noted that the modal group is 99%–100% as the accuracy of most of the prediction fall in this group.

**Fig. 6.** Frequency distribution in accuracy of the predictive thermal model for the complete data set in the Celsius scale.

The accuracy and estimation time of the proposed model are also compared for ITC'02 benchmark circuits d695 placed in a 4-die 3D IC as in [18] and p93791 placed in 3-die 3D IC. Table 3 shows the temperature of the 3D ICs for various active cores obtained from the proposed model and compared with the temperature obtained from Hotspot and Comsol. It is observed from Table 3 that the temperatures estimated by the proposed model are similar to the mathematical models and the CPU execution time is drastically reduced.

In the proposed test scheduling algorithm, cores are added to a session on the basis of whether the temperature exceeds the thermal safety limit. For this reason, the classification accuracy is also important. Fig. 7 plots the predicted temperature versus the actual temperature of each random 3D IC in the data set for thermal limit of 90°C. The blue line has the ideal slope of one and points on the line represent 3D ICs whose predicted and actual temperature are equal. Hence, points clustered close to this line yields higher classification accuracy. The green dots in Fig. 7 represents correct prediction classification, where both the predicted and actual temperature are either below or above the thermal safety limit. While, wrong classifications are depicted by red dots. The confusion matrix is inset in the figure, where it is observed that the total classification accuracy is 98.97%.

**Fig. 7.** Classification accuracy of the predictive thermal model of the entire data set at thermal safety limit and ambient temperature of 90°C and 27°C respectively.

The predictive thermal model must perform well over a wide range of thermal safety limit because the thermal safety limit varies over different technology nodes. Table 4 shows the confusion matrix data and classification accuracy, recall and precision of the model over the entire data set for various thermal safety limit ranging from 60°C to 130°C. Both actual and predicted temperatures being safe are considered the true positive cases for classification. Table 4 shows that accuracy, recall and precision of the proposed ML-based model is quite high for a wide range of thermal safety limits. This results in misclassification of core combinations (safe core combinations being classified as unsafe or vice-versa) to be quite rare in the test scheduling process.

6.2. Test scheduling

Test scheduling results of three ITC'02 benchmark circuits are presented in this sub-section, which are placed separately into three 3D ICs. Benchmark SoC d695 is placed in four dies as in [18]. Benchmark SoCs p22810 and p93791 are both placed in 3D ICs containing three dies each. The cores are distributed among the dies in a way such that the power dissipation and area of each die is balanced. The power dissipation is estimated as the average over windows of 400 clock cycles at a base test clock frequency of 120 MHz at 45 nm technology node [31]. The test time of the cores for various TAM widths is calculated as presented by Marinissen et al. [28].

Test scheduling results of d695, p22810 and p93791 are presented in Table 5, Table 6, and Table 7, respectively, and compared with some existing works in literature [2,18–21]. All of the above-mentioned tables are broadly divided into two parts — Existing and Proposed. The first part shows the results present in the literature, whereas the results in the second part pertain to this work. The results of this work are further sub-divided into two sections — one presenting the details of test schedule with minimum clock cycles (CC), while the other presents the schedule with minimum absolute time (in milli-seconds).

Table 3

Temperature and CPU runtime of the predictive thermal model compared with that of Hotspot [3] and Comsol [4] for various combination of cores of d695 and p93791 being active.

3D IC	Active cores	Hotspot [3]		Comsol [4]		Proposed			
		Temp. (°C)	CPU time (s)	Temp. (°C)	CPU time (Min)	Temp. (°C)	CPU time (s)	Accuracy w.r.t Hotspot (%)	Accuracy w.r.t Comsol (%)
d695	1, 2, 3	45.93	7.54	48.13	25	46.97	0.0104	97.73	97.59
	1, 2, 3, 4	49.08	7.747	52.18	24	50.11	0.0091	97.9	96.03
	5, 6	61.41	9.087	65.02	24	64.88	0.0077	94.34	99.78
	5, 6, 9	76.51	10.448	84.27	24	80.14	0.0091	95.26	95.1
	3, 4, 5, 8	57.45	8.403	63.43	23	59.5	0.0101	96.43	93.8
	1, 3, 4, 8, 10	50.18	8.109	53.5	25	51.37	0.0098	97.63	96.02
	2, 5, 6, 7, 9	94.8	11.962	102.87	24	98.34	0.0077	96.27	95.59
p93791	12, 24	44.47	9.265	47.12	23	46.48	0.0122	95.48	98.64
	10,19,24, 25	58.02	10.865	61.03	24	58.54	0.0106	99.1	95.92
	4, 8, 21, 22, 26	39.02	8.92	42.12	24	38.62	0.0084	98.97	91.69
	1, 14, 21, 28, 31	49.72	9.7	52.9	25	51.11	0.109	97.2	96.61
	1, 6, 11, 20, 27	75.14	11.959	79.06	26	78.47	0.0115	95.57	99.25
	2, 3, 15, 16, 28, 31	37.84	9.286	39.04	24	35.54	0.101	93.92	91.03

Table 4

True Negative (TN), True Positive (TP), False Negative (FN), False Positive (FP), Accuracy, Recall and Precision of the predictive thermal model over entire data set for various thermal safety limit (T_{limit}).

Safe thermal limit T_{limit} (°C)	TN (%)	TP (%)	FN (%)	FP (%)	Accuracy (%)	Recall (%)	Precision (%)
60	62.78	35.98	0.068	1.169	98.76	99.81	96.85
70	52.54	46.51	0.108	0.835	99.06	99.77	98.24
80	42.86	56.33	0.147	0.658	99.19	99.74	98.84
90	33.68	65.29	0.177	0.854	98.97	99.73	98.71
100	23.43	75.82	0.147	0.599	99.25	99.81	99.22
110	16.84	82.72	0.118	0.314	99.57	99.86	99.62
120	11.83	87.73	0.157	0.275	99.57	99.82	99.69
130	7.81	91.87	0.098	0.226	99.68	99.89	99.75

It also presents the $fScale$ value for each schedule - a factor by which the base test clock can be scaled. CPU Execution time and percentage improvement in test time over existing works are also shown. It is observed from the above-mentioned tables that the test time of the schedules generated by the proposed algorithm is much lower than existing schedules in most of the cases. The main contributing factor behind it, is possibly the incorporation of the test clock frequency scaling in the scheduling algorithm. Introduction of clock frequency scaling better optimizes the schedule both in terms of clock cycles and absolute time. Tweaking of the test clock frequency makes the power dissipation of the cores malleable, which in turn helps to generate a better optimized test schedule.

The thermal evaluation technique used in this work is based on machine learning. So, there are bound to be some false positive cases where the predicted temperature is below the thermal safety limit (T_{limit}), but in actual it is not. This necessitates a final and accurate thermal estimation of the generated test schedule. This must be done to ensure absolute thermal safety and high confidence in the test procedure. In such cases, the test clock frequency is adjusted by changing the $fScale$ value according to Eq. (9) to bring the actual temperature below T_{limit} . One such case that we encountered in this work is for p93791 with TAM width of 64. Originally, the actual temperature of the generated schedule was found to be 91.1 °C at $fScale$ value of 1.85. The thermal safety of this schedule is ensured by adjusting the

$fScale$ value to 1.8, which brought the actual temperature down to 89.37 °C.

The thermal safety limit also plays a vital role in determining test time and is dependent on the technology. Commercially available chips generally have a recommended safety limit of 70 °C, whereas the high-end military and medical applications have a limit of 125 °C. A high thermal safety limit facilitates more cores to be tested in parallel and hence, reduced test time. Incorporating the test clock frequency can also produce optimized results for different thermal safety limits, thus eliminating the need of new test scheduling algorithms for different thermal limits. Fig. 8 shows the change in absolute test time of d695, p22810 and p93791 for various thermal safety limits and TAM widths.

The number of TSVs required for testing is also important as TSVs add to the test cost. Fig. 9 shows the number of TSVs required by the proposed test scheduling algorithm for various 3D IC benchmarks. It is observed from the results that the number of TSVs required by our testing scheme is comparable with existing test schemes [18,20,21].

7. Conclusion

Three-dimensional Integrated Circuit (3D IC) is a new technology that alleviates many of the problems faced by conventional 2D ICs. However, it also faces some critical challenges like thermal management and testing. This work proposes a machine learning based thermal

Table 5

Test time and temperature of d695 for various TAM width compared with existing works.

TAM	Existing								Proposed												CPU Exec. time (s)	Improvement in test time (%)	
	Chatterjee et al. [2]		Banerjee et al. [18]		Chandran et al. [19]		Wu et al. [21]		Minimum in clock cycles (CC)						Minimum in absolute time (mS)								
									Test time			Temperature (°C)			Test time			Temperature (°C)					
	CC	mS	CC	mS	CC	mS	CC	mS	CC	fScale	mS	Hotspot	Proposed	CC	fScale	mS	Hotspot	Proposed					
16	44901	0.374	42833	0.357	44027	0.367	44225	0.369	43233	1.4	0.257	83.82	88.21	44253	2.55	0.145	84.69	89.03	6.88	-0.9	59.4		
24	31758	0.265	31569	0.263	32542	0.271	31070	0.259	28775	1.75	0.137	84.97	88.46	32564	2.55	0.106	86.32	89.96	5.21	7.4	59.1		
32	24673	0.206	23297	0.194	24133	0.201	23944	0.2	22217	1.65	0.112	83.77	89.06	24485	2.4	0.085	86.29	89.78	6.09	4.6	56.2		
40	20050	0.167	18862	0.157	19718	0.164	21070	0.176	18526	1.15	0.134	86.3	88.92	24021	2.55	0.079	86.32	89.96	6.14	1.8	49.7		
48	16936	0.141	18333	0.153	18837	0.157	21070	0.176	15069	1.35	0.093	83.86	88.11	20276	2.4	0.07	86.29	89.62	6.06	11	50.3		
56	15737	0.131	15555	0.13	15237	0.13	21070	0.176	12717	1.0	0.106	84.23	87.62	19846	2.4	0.069	86.29	89.62	6.43	16.5	46.9		
64	15713	0.131	13732	0.114	13816	0.115	21070	0.176	11274	0.7	0.134	86.8	89.0	19846	2.4	0.069	86.29	89.62	6.16	17.9	39.5		

Table 6

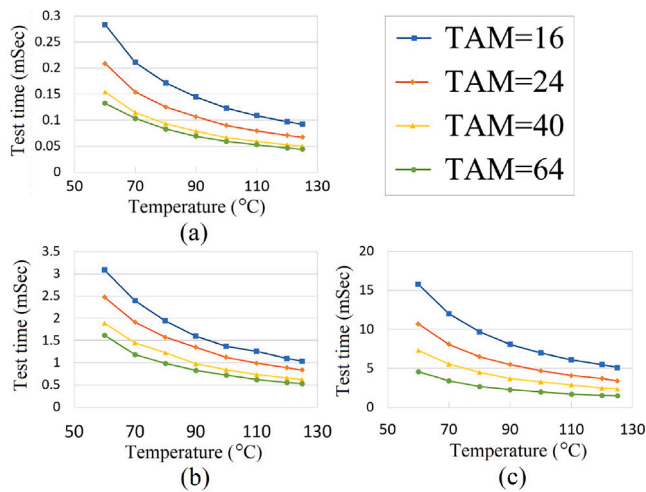
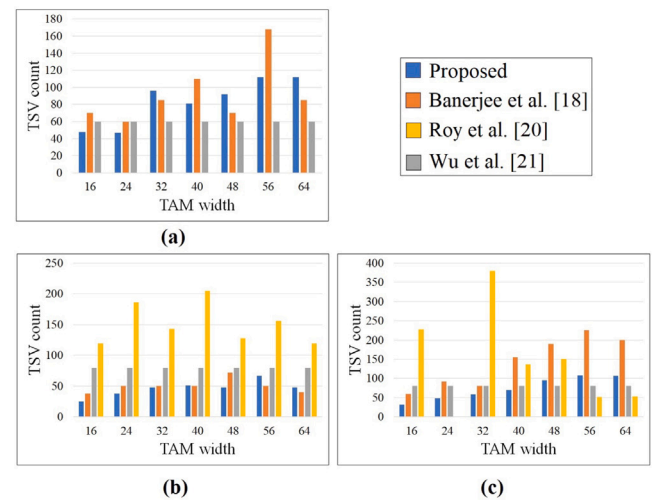
Test time and temperature of p22810 for various TAM width compared with existing works.

TAM	Existing								Proposed												CPU Exec. time (s)	Improvement in test time (%)		
	Chatterjee et al. [2]		Banerjee et al. [18]		Chandran et al. [19]		Roy et al. [20]		Minimum in clock cycles (CC)						Minimum in absolute time (mS)									
									Test time			Temperature (°C)			Test time			Temperature (°C)						
	CC	mS	CC	mS	CC	mS	CC	mS	CC	fScale	mS	Hotspot	Proposed	CC	fScale	mS	Hotspot	Proposed						
16	461119	3.84	485728	4.05	469487	3.91	459932	3.83	460624	2.25	1.7	89.62	89.15	471648	2.45	1.6	89.79	89.97	21.4	-0.2	58.2			
24	331194	2.76	371217	3.09	376819	3.14	343608	2.86	317972	1.35	2.0	88.67	88.7	406368	2.5	1.35	88.9	89.89	21.7	3.9	56.5			
32	243498	2.03	252887	2.11	270592	2.25	243303	2.03	240383	1.0	2.0	89.37	87.23	291717	2.2	1.11	89.43	89.39	18.6	1.2	45.3			
40	217373	1.81	191580	1.6	240542	2.0	192193	1.6	203479	1.2	1.4	88.65	87.85	258685	2.2	0.98	89.14	89.86	22.2	-6.2	38.7			
48	177929	1.48	195269	1.63	179755	1.5	176251	1.47	168065	0.95	1.5	89.12	87.19	188699	1.8	0.87	88.42	89.86	20.0	4.6	40.8			
56	159792	1.33	165352	1.38	157402	1.31	155884	1.3	150172	0.8	1.6	89.96	88.5	176314	1.7	0.86	88.35	89.73	18.8	3.7	33.8			
64	158999	1.32	142428	1.19	146630	1.22	134820	1.12	131747	1.2	0.9	89.98	89.79	179852	1.8	0.83	88.52	89.86	18.9	2.3	25.9			

Table 7

Test time and temperature of p93791 for various TAM width compared with existing works.

TAM	Existing								Proposed														
	Chatterjee et al. [2]		Banerjee et al. [18]		Wu et al. [21]		Roy et al. [20]		Minimum in clock cycles (CC)						Minimum in absolute time (mS)						CPU	Improvement	
									Test time			Temperature (°C)			Test time			Temperature (°C)			Exec. time	in test time (%)	
	CC	mS	CC	mS	CC	mS	CC	mS	CC	fScale	mS	Hotspot	Proposed	CC	fScale	mS	Hotspot	Proposed	(s)	CC	mS		
16	1850137	15.42	1856606	15.47	1869200	15.58	1796738	14.97	1803620	1.45	10.4	84.97	89.74	1803800	1.85	8.1	82.39	88.67	34.3	−0.4	45.9		
24	1227781	10.23	1256380	10.47	1294512	10.79	–	–	1208516	1.55	6.5	84.3	89.45	1210490	1.85	5.5	82.39	88.67	39.7	1.6	46.2		
32	981770	8.18	1108271	9.24	966752	8.06	1060440	8.84	940398	1.25	6.3	87.36	88.4	973699	1.85	4.4	87.84	89.47	23.2	2.7	45.4		
40	747407	6.23	1103105	9.19	778480	6.49	1033275	8.61	740619	1.15	5.4	87.06	89.89	763545	1.7	3.7	89.11	89.92	28.1	0.9	40.6		
48	618598	5.15	634441	5.29	766270	6.39	624851	5.21	618136	1.55	3.3	84.3	89.45	622043	1.85	2.8	83.23	89.51	52.2	0.1	45.6		
56	584972	4.87	604257	5.03	685314	5.71	575251	4.79	530986	1.85	2.4	89.57	89.47	530986	1.85	2.4	89.57	89.47	33.9	7.7	49.9		
64	488665	4.07	514652	4.29	662198	5.52	475083	3.96	470984	1.1	3.6	89.45	87.86	506345	1.8	2.3	89.37	89.74	32.4	0.9	41.9		

**Fig. 8.** Variation in absolute test time (milli-second) for different thermal safety limit of (a) d695, (b) p22810, and (c) p93791.**Fig. 9.** Variation in the total TSV count using the proposed test algorithm compared with other works for (a) d695, (b) p22810, and (c) p93791.

estimation scheme for 3D ICs that drastically reduces the thermal estimation time without sacrificing much in accuracy. The thermal estimation model has been tested with several temperature safety limits, and experimental results show a high degree of classification accuracy, recall, and precision. The predictive thermal model has been used to present a new test scheduling algorithm for 3D ICs. The generated schedules are heuristic in nature, session-based and incorporate test clock frequency to minimize test time. Several ITC'02 benchmarks [32] have been tested with the proposed algorithms, and most cases show improvement in test time over existing works. A maximum reduction in test time of around 17% (in terms of clock cycles) and around 60% (in absolute time) has been observed.

CRedit authorship contribution statement

Subhajit Chatterjee: Conceptualization, Visualization, Methodology, Software, Data curation, Writing – original draft. **Surajit Kumar Roy:** Writing – review & editing, Investigation, Supervision. **Chandan Giri:** Supervision. **Hafizur Rahaman:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.mejo.2022.105535>.

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